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Modelling patient flow in an emergency department to better understand demand management strategies

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This paper presents a simulation model of the emergency department (ED) at a major UK hospital, which like most other EDs, is struggling to meet the key ED performance target to admit or discharge 95% of patients within 4 h of arrival. Only marginal improvements have been evidenced from recent investigations piloted as attempts to improve cost, quality and throughput in the ED, leading to increased enthusiasm from management to explore the potential of simulation to expedite patient flow. The objective of this project was to create a discrete event simulation model of the ED to improve understanding of the current system, identify shortcomings of previously piloted interventions and assess the impact of alternative initiatives. The project has served as an excellent means to promote simulation as an effective tool to design ED services. Its success to generate outputs above those anticipated by management has accelerated the development of a multidisciplinary Flow Team, which meets regularly at the hospital to review improvements and explore further proposals to improve patient flow.

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1. Introduction

Facing ever increasing pressures from rising public expectations and an ageing population requiring increasing levels of care and resource, the impact of dwindling budgets and growing demand is felt across the health service. Emergency departments (EDs) are however often the first to exhibit strains when demand requirements exceed capacity (NHS Confederation, 2012; Saghaian *et al*, 2014). Intended to deal with emergencies requiring urgent medical intervention, ED's necessary visibility has led to them becoming a catch-all service for patients unable or unwilling to access alternative care, resulting in numerous reports of overcrowded waiting rooms with long stays over the last decade (see Derlet and Richards, 2000; Rondeau and Francescutti, 2005; Magid *et al*, 2009; Bernstein *et al*, 2008, and references within). Since the functioning of each ED is intrinsically linked to the effectiveness of every other possible contact point for each patient, there is clearly no single 'cure' for EDs. This paper instead considers the contribution of selected strategies which might help bring the patient and ED provider together more expeditiously (Wiler *et al*, 2010).

The most recognised quality of care measurement for EDs in the UK is their compliance against the 4 h national performance target, which states that 95% of patients should be admitted or discharged within 4 h of arriving at the ED. In attempts to streamline the front-end processing of ED patients and meet this target, several interventions such as immediate bedding, advance triage, smart cards and tracking systems

have been explored, which become crucial during periods of full capacity (Wiler *et al*, 2010). However, despite such grand efforts to improve patient flow, only 88% of ED patients in Wales and 93% in England received appropriate treatment within the targeted time in 2013/14 (see NHS England, 2014; StatsWales, 2014). There is therefore a great call for alternative methods to improve the design and delivery of quality healthcare. One such initiative which has recently been launched in Wales is the Patient Flow Programme (Health Foundation, 2012). This has prompted the set up of multidisciplinary Patient Flow teams throughout the country which meet together on a weekly basis to identify delays in their respective systems and investigate ways of achieving smoother patient flow. Taking the emphasis away from the manmade 4 h target, the programme focusses on exploring the relationship between patient flow, costs and outcomes and developing ways in which capacity can be better matched to demand.

This paper provides strong motivation for the use of Discrete Event Simulation (DES) to promote emergency patient flow, focussing in particular on the operations at the Royal Gwent Hospital. Located in Aneurin Bevan University Health Board (ABUHB) in South Wales, the hospital serves a population of around 350,000 and offers a comprehensive range of hospital services. In an attempt to effectively respond to an ambitious Welsh Government agenda to improve the quality of care it provides within a constrained budget, ABUHB established the Aneurin Bevan Continuous Improvement centre (ABCI) in 2012, with the purpose of integrating existing improvement approaches across the Health Board (Aneurin Bevan Health Board, 2012). A key part of the

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improvement centre is an embedded Mathematical Modelling Unit, which worked closely with the Unscheduled Care division in this project. Prior to the introduction of modelling techniques to the hospital, several innovative demand management strategies were piloted at the ED, but very few proved sustainable or successfully evidenced anticipated improvements in patient flow and congestion levels. The primary objective of this project was thus to determine the underlying causes for the shortcomings of the pilots and recommend where future efforts could best be placed to realise the greatest improvements. Through demonstrating the ability of simulation models to add in sufficient complexity to accurately reflect the operations of the ED as they stand and carefully adjusting parameter values to reveal gains that could be achieved from alternative configurations, both managers and clinical staff at the department have been able to appreciate the benefits of modelling to test out proposed interventions prior to implementation. Indeed, modelling is starting to be viewed as a core component of the planning process at the Health Board.

One of the key ways in which the hospital has attempted to decongest the ED was through the establishment of the clinical decisions unit (CDU) housing 12 beds in January 2013. The unit aims to promote rapid assessment and turnaround of patients who present relatively straightforward medical conditions with expected length of stays (LoS) between 4 and 24 h. Its underlying intention is to essentially pull suitable medical patients out of the ED as early as possible in their stay, to treat them efficiently and expedite their discharge; thus allowing them to return to their normal lifestyles as quickly as possible. In this way, the unit was anticipated to reduce default admissions to inpatient beds, decongest the Majors area of ED, improve the patient experience, reduce waiting times and therefore impact positively on ED performance and quality indicators (Mizen *et al.*, 2013). There has been significant literature about the effectiveness of the so-called fast streaming (see Counselman *et al.*, 2000; Cooke *et al.*, 2002), and feedback from clinicians suggests that this strategy has been somewhat successful. Nevertheless there are still frequent reports of bottlenecks similar to those seen before, meaning that patients are often being 'pushed' into CDU, in place of being 'pulled' from ED. This project was initiated 6 months after the introduction of the new layout in the ED. Since the expected reduction in average length of stay was not being achieved, the objective of this work was to investigate why and also look at other 'what if' scenarios in order to aid the Flow Team on alternative measures to best achieve reduced average LoS.

Other strategies that have been piloted at the hospital to promote patient flow include a pre-hospital streaming process, whereby the Welsh Ambulance Service Trust (WAST) contacted the hospital in advance to arrange direct admission of suitable medical patients to the CDU (effectively bypassing ED) and also a small GP streaming trial, in which GP referrals were intercepted by a consultant who aimed to redirect or reschedule attendances. Statistical analysis of the trials revealed that the

streaming was successful in reducing waiting times in the ED but limitations in resources and clinical backing limited the extent to which the streaming was administered. In fact, the GP streaming trial was only carried out by a single Acute Physician Consultant over a period of two days! It was extremely unlikely to be reinstated without overwhelming evidence to support its execution. The client thus specifically requested for scaled-up versions of the trials to be considered as 'what-if' scenarios in the model (see Section 3.3, to assess potential impacts associated with their implementation.

The remainder of this paper is organised as follows. In Section 2 we review related literature and in Section 3 we outline the process followed to construct an accurate replication of the ED. The use of the model to explore what-if scenarios is described in Section 3.3, before conclusions and recommendations for future research are ultimately given in Section 4.

2. Literature review

There are broadly two complementary courses of action to improve ED performance: (i) reducing attendance at the ED, and (ii) improving patient flow through ED. This study has focussed on the latter, but these must be considered in the context of the former. Van Woerden *et al.* (2013) recently reviewed the pressures on acute care in Wales, emphasising that beyond self-care, there is a need to provide sufficient support for patients whose conditions are manageable in the community but access emergency care instead. This is a complex issue involving many different disciplines. Discussions with staff at the department indicated that the implementation of these techniques in an efficient way is difficult for a number of reasons, not least the variability in geographical factors and patient need, hence this project evolved as a study to investigate feasible methods to expedite the flow of patients through the ED.

The highly complex non-terminating nature of hospital services with limited specialised resources and high levels of stochastic demand has long attracted the attention of operational researchers in several areas, including forecasting, capacity planning, queueing and resource allocation (Brailsford, 2005). It is therefore unsurprising that the literature devoted to operational research (OR) methods improving the performance of EDs is vast. Whilst ED managers traditionally attempted to wrestle with the challenge to deploy the right number of staff members with the right skills in order to meet an uncertain, time-varying demand for service by hand; simulation is now widely accepted as an effective tool for process improvement (Sinreich and Marmor, 2005; Duguay and Chetouane, 2007). As hospital managers are increasingly being pressurised to reduce costs, simulation has the potential to offer many benefits through revealing the underlying interactions and helping stakeholders appreciate the complexity of the system and variability within each resource type (Ashton *et al.*, 2005).

However, successful implementation of simulation models is mixed. Bowers *et al* (2009) suggest that both potential clients and analysts need to establish realistic expectations before starting the model building process and appreciate the challenges of simulation modelling in a healthcare environment.

After reviewing more than 300 papers on the topic, Saghafian *et al* (2014) concluded that the ability of Discrete Event Simulation (DES) to add in a great level of complexity makes it an ideal candidate for virtually every aspect of the ED that impacts patient flow. With an emphasis on patient flow, they identified a number of papers that promote the use of simulation in the ED (Fone *et al*, 2004; White, 2005; Gunal and Pidd, 2010). Accurate simulations of ED flow have been constructed in studies such as those by Samaha *et al* (2003) (which dispelled a common fallacy that more beds improve patient flow), Diefenbach and Kozan (2011) and Wang (2013); and over 90 papers that used simulation to examine the impact of patient flow in ED on overcrowding have been reviewed in Paul *et al* (2010). Moreover, the ability of patient streaming to offer improvements in ambulance turn away rates, patient LoS and up to 50% reductions in waiting times have been demonstrated with simulations (see Kelly *et al*, 2007; Peck, 2009; Saghafian *et al*, 2012, 2014; Ahmed and Alkhamis, 2009). Fast tracking lower acuity patients in the ED has synchronously been shown to be successful in Williams (2006), Roche and Cochran (2007) and Welch (2009).

When modelling the ED, it is nevertheless crucial to remember that the flow of patients through the department is not only affected by the availability of resources within the ED, but also by systems influencing the flow of patients out of the department (Diefenbach and Kozan, 2011). A recent study by Baboolal *et al* (2012) has broken away from the traditional modelling approach, instead building a perfect world model and asking why reality does not reflect this. Detailed information was gathered from an ED and a model reflecting the patient flow was built. Key parameters were altered, including bed numbers and staffing levels. This provided a clear indication of the point at which additional resources stop adding significant value by reducing waiting times. By assigning a cost associated with each trolley, the team was able to offer a conservative estimate of the costs and savings that would result from their scenarios which has been invaluable in informing future policy. In another study, Komashie and Mousavi (2005) compared a number of Key Performance Indicators (KPIs: LoS, queuing time and resource utilisation) from 5 different scenarios that varied parameters such as number of beds or doctors. They noted that LoS in the Minors department was much more sensitive to staff numbers than admission delays or number of cubicles, whereas the most significant drop in LoS for Majors patients was when ‘admission blockage’ was removed.

Whilst this literature review has identified a wide range of models developed to evaluate potential approaches to improve cost, quality and throughput efficiencies in the ED, a recurring problem has been the lack of real-world implementation. A study by Day *et al* (2013) goes some way to addressing the gap

between academic research and implementation in a quasi-experimental trial of an additional healthcare provider in triage. Bell *et al* (2006) note that some refinement is often needed for the model recommendations to retain a focus on the needs of the patient.

In this study, we have attempted to bridge the gap between consultation and implementation and contribute to the existing literature by:

- Not only involving front line staff in the modelling exercise from day one, but regularly discussing adaptations of the model, impacts to date and combining OR techniques with Six Sigma methods (e.g. ‘Plan-Study-Do-Act’ cycles and process maps) in weekly multidisciplinary flow meetings. These meetings are typically attended by a team of clinicians, managers, analysts and improvement leads who meet to debate solutions that could potentially eliminate waste and improve the quality of care;
- Responding to the calls in Eitel *et al* (2010) and Bowers *et al* (2009) to identify and target a few key stakeholders to become champions for the change initiative. Also appreciating the particular challenges of simulation in a healthcare environment with the healthcare team and establishing realistic expectations from the outset;
- Using modelling to highlight problems with the way previously trialled solutions had been implemented, through revealing the data underpinning their systems not visible to the naked eye, rather than simply investigating future proposals;
- Considering how the behaviour of staff working at related services (e.g. GP surgeries, ambulance services) could be changed to influence the flow of patients into the hospital, including bypassing ED in ‘what-if’ scenarios;
- Leading the way for other Health Boards and Trusts to integrate modelling and simulation into their planning through demonstrating the ability of simulation modelling to quickly pinpoint underlying bottlenecks in the system and predict expected measurable benefits of alternative configurations. Hospital managers and clinical staff have become convinced of the value of this approach at ABUHB and in agreement that modelling is crucial to planning future services in the right way.

Taking everything into account, the project has created a growing sense of ownership amongst staff working at the hospital ED and established an excellent environment for exploring the use of simulation as an embedded support tool for healthcare design.

3. Methodology

As demonstrated in Section 2, simulation has been proven as an effective tool for process improvement in numerous ED studies. The general steps that should be followed in a typical simulation study was first formalised by Banks *et al* (1999)

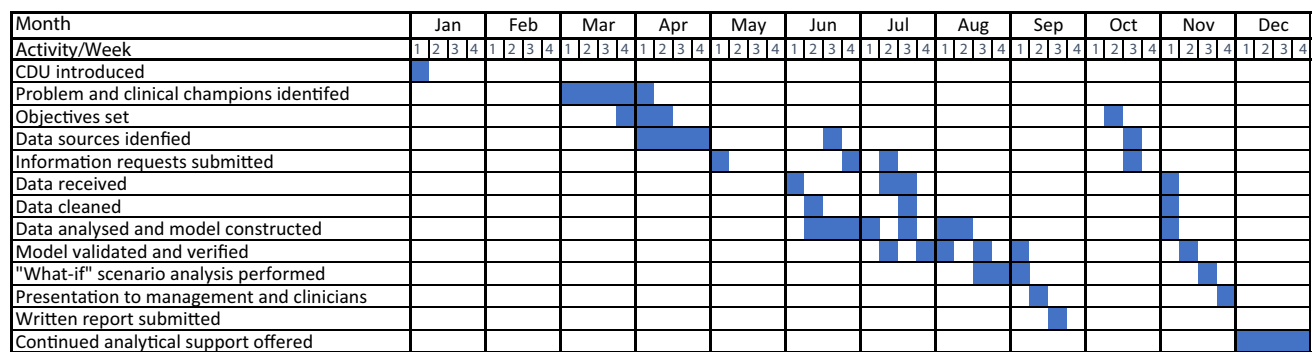


Figure 1 Key stages in simulation study.

and its practical accuracy has been tested in several studies since. The steps followed in this study have affirmed that the general process is still relevant today, provided there is flexibility as to the exact ordering of the steps in an iterative process. An indication of the times that were necessary to identify stakeholders and each of the steps in this specific study are provided in the Gantt chart in Figure 1 below.

The Gantt chart demonstrates the importance of regularly meeting with stakeholders in order to capture all of the process interdependencies (which was only successfully achieved after several model iterations) and the requirement to quickly react to feedback from key decision makers. Although significant time was invested in defining the project objectives at the start of this project, the findings of some of the preliminary studies unearthed additional information which meant that it was necessary to later revisit and repost some of the original questions. Although time-consuming, the iterative development process was invaluable in gaining the trust and confidence of the clinicians and wider management team. It is however worth noting that the modeller in this project benefitted from being viewed as a ‘colleague’ (from within the Health Board’s Mathematical Unit) and thus able to access to a wide range of data at speed from two of the Health Board’s Patient Administration Systems (namely ‘Symphony’ in the case of ED data and ‘Myrddin’ for inpatient data) quicker than an external modeller might expect. Considerable time was nevertheless required between the emergence of the project findings and the start of the change management process in the hospital. In this project, physical changes began to take effect in the following January and are still ongoing today (see Section 3.8).

Further information on the construction of the simulation model itself follows below and a summary of the software packages that were used in this project is provided in Appendix A. Whilst highlighting that a simulation project critically requires more than one software package to succeed, the software list also includes suggestions of alternative options where appropriate, to give project managers wishing to undertake simulation studies a realistic idea of the software investments necessary.

3.1. Model development

To accurately represent this highly variable and complex system whilst maintaining focus on patient welfare, a DES model was constructed in SIMUL8. Three main departments (Majors, Minors and Paeds) were included in the model. Patients’ routes through the system were determined by their triage category, which was assigned according to their method of arrival (ambulance arrivals tended to be more severe than walk-in patients for example). A number of possible exit routes were identified, including discharge, referral, transfer and admission. Two wards were included in the model; the medial assessment unit (MAU) and the short stay unit (SSU), incorporating the waiting time to receive a ward bed.

Given the multi-faceted nature of the ED and a requirement for communicability to healthcare managers, the level of detail of the simulation model was given careful consideration and discussed at weekly multidisciplinary flow meetings. A patient flow map (Figure 2), detailing all possible pathways through the ED was created in a consultation process with clinical staff who confirmed its accuracy and suggested areas for investigation. This facilitated a rounded understanding of system behaviour which was used to verify the model, following guidelines provided in Sargent (2013). Clinical champions were also identified at the weekly meetings and given responsibility for directing the development of the model, providing expert opinion and revealing further insights (e.g. about extreme and unusual scenarios). These individuals were responsible for ensuring the recommendations suggested were both realistic and achievable and ultimately implemented within the workplace.

The following sections describe the model building processes undertaken to provide a computerised model of the ED at the hospital. For this purpose, 18 months of historic activity (January 2012–June 2013) was obtained from the ED computerised recording system, Symphony and complimented with expert opinions elicited from the clinical champions in a series of interviews. In the majority of cases their opinions agreed, but in cases where the difference in the estimates given was non-negligible, we asked them to debate their estimates

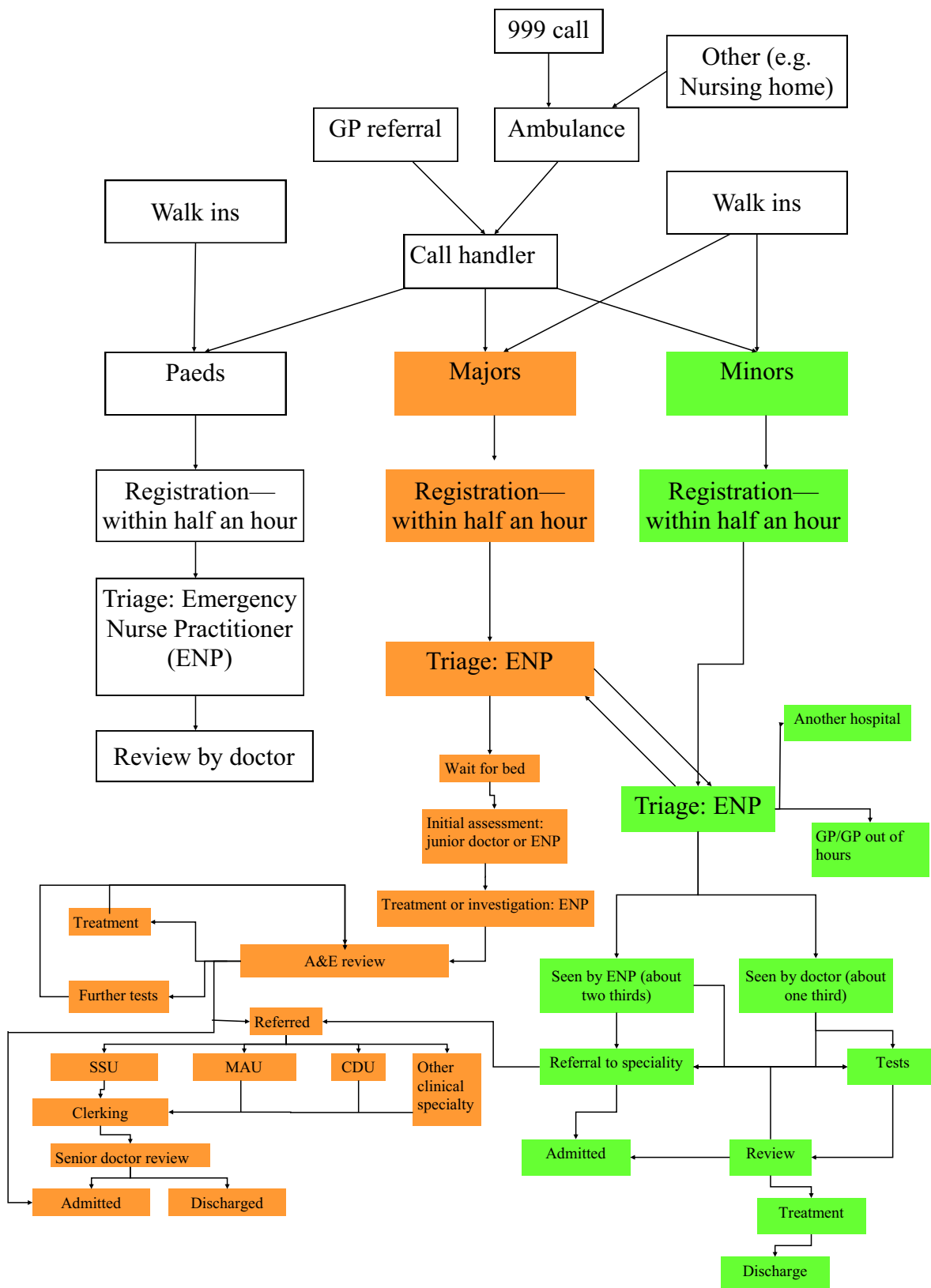


Figure 2 Patient pathways through the ED.

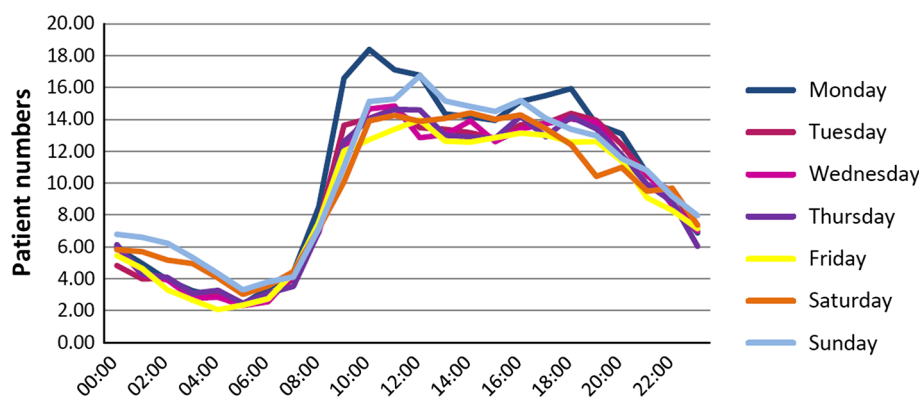


Figure 3 Average hourly presentations in 2012, by day of week.

until they were able to provide time ranges without conflict. The last 6 months of data (January - June 2013) were held back to assess the impact of the CDU on waiting time and breaches, since it had been established. In line with requests from hospital staff to make the simulation model as user friendly and flexible as possible, an Excel interface was also created to enable planners with minimal technical expertise the opportunity to adjust parameter values and explore other scenarios.

Statistical analysis of the data found that many factors influenced attendances at the ED, including the day of the week and temperature. In addition, there was significant variability between patients which resulted in highly variable LoS. Indeed age, a key factor identified as influencing LoS, accounted for only 8% of the variation. Attendance numbers were found not to correspond to LoS in the ED. The effect of the 4 h target on LoS distribution was striking, with a vertiginous drop in numbers discharged just after 240 min.

3.1.1. Arrival characteristics Arrivals to the ED were found to be highly stochastic, well suited to a non-stationary Poisson process with a time-dependent rate factor and influenced by external factors such as day of the week and temperature. The pattern in daily attendances can be seen in Figure 3; peaking mid-morning, levelling off over lunch time and decreasing throughout the evening. Sunday and Monday show some similarities, as do Friday and Saturday, and Tuesday, Wednesday and Thursday. The mornings of Saturday and Sunday are significantly higher than the others. This clustering of days is widely mentioned in the literature, though the groupings vary. Indeed, Sahu *et al* (2014) found that admissions to another neighbouring hospital also peaked on Mondays, but grouped demand patterns on Tuesday–Friday and Saturday–Sunday together.

Historic data were used to identify a comprehensive arrival pattern for patients arriving by different modes (walk-in, GP referral, nursing home, etc.), time of day, day of week and month of year. The ability of the model to consider daily and monthly effects separately allowed the modelling team to pick

up on subtleties such as GPs being closed at the weekend or older people being more vulnerable to colder temperatures resulting in more significant increases in attendance from nursing homes in the winter. Ultimately, arrival distributions were fitted to patient groups for statistically similar groups of months and weekdays (determined with ANOVA/Kruskal–Wallis tests) e.g. walk-in, months 1–6, days Sunday–Monday, hours 13–16, using the **R** package *fitdist*).

3.1.2. Profiling Patients arriving at the ED were profiled according to their triage category (1–6), assigned upon entry to the system according to previously observed proportions of categories within each source of service request. Visual Logic then used this category label at the call handling point to route patients between Majors, Minors and Paediatric departments. This method enabled category proportions to reflect the characteristics of the sources, for example self-referral ‘walk-in’ patients generally had a higher category than those brought in by ambulance. Further profiling was explored (for example age and type of injury) but both data constraints and complexity issues meant that it was not implemented, though this is suggested for further research.

3.1.3. Activities Activities in the model included all significant events in a patient’s treatment as well as some additional ‘dummy’ activities to reflect real systems. Activity times were specified for each activity which in most cases was a distribution with parameters specified by the user in the Excel interface. The time taken to perform each activity was in most cases estimated, informed by staff feedback and visits to the department. Six main activities associated with a stay in the department were identified:

1. *Registration* Performed in waiting area or on a trolley; may occur simultaneously to triage.
2. *Triage* An initial assessment of patient is made using the national Manchester Triage tool (see, Cooke and Jinks, 1999) to clinically prioritise patients.

3. *Clerking* Patient is seen by a doctor or an Emergency Nurse Practitioner in the case of minor injury patients.
4. *X-ray* Sometimes a necessary investigation (indicated at the point of assessment).
5. *Review* Doctor will review patient post-investigations, which may require senior review.
6. *Treatment* An ambiguous term to include a wide range of possible actions e.g. plastering of broken bone/rehydration with a saline drip.

3.1.4. Wards Despite the model concentrating on the ED (i.e. *before* admission), a small number of relevant wards were also modelled. This was necessary both for the later exploration of ‘what-if’ scenarios involving streaming directly to these wards, and to incorporate the effect of downstream ‘bed-blocking’ without adding too much complexity to the model. These included the MAU (an assessment unit for medical patients, consisting of 15 beds) and the SSU (a 9 bed unit intended for patients with a clearly defined treatment plan, mainly requiring observation and treatment, with an expected discharge within 24 h).

3.1.5. Routing The percentage of patients following different routes was deduced from limited information available in the data and supported by observations. In most cases, queues allowed lower category patients to be prioritised. Each of the duplicated activities was served by a single queue with no limit. The lack of limit does not reflect reality (it is unrealistic to have 100 Majors patients in the waiting room for example) but highlights extreme situations that would trigger actions (such as increased escalation levels) designed to counteract peaks in demand through the system.

3.1.6. Resources Once the structure was in place, the constraints of staff and bed availability were introduced as resources. The staffing grades and shift patterns were informed from meeting with an HR manager at the hospital, and their responsibilities by clinician interviews. The main clinicians are (in increasing order of responsibility): Qualified nurse, Advanced Nurse Practitioner, Emergency Nurse Practitioner (ENP), Junior Doctor (SHO), Middle Grade (SpR), Associate Specialist, Consultant. For each staff type, a number of shift slots were identified from example staffing rotas supplied by staff managers. These shift patterns were created and assigned based on an average week. Staff members were able to finish their task before leaving at the end of their shift, which was felt to more accurately reflect staff behaviour in the department. Additional staff in the model included receptionists and staff dedicated to ‘peripheral wards’ such as the SSU.

Pooled resources were created where appropriate and staff availability was reduced from 100% to more accurately reflect real behaviour in accordance with the findings of a quality

improvement project recently undertaken by the HR manager and substantiated by consultants responsible for managing the rota on the ground. For example, nurse availability was reduced to 85% to account for breaks and administration tasks not included in the model, ENP availability was set at 90%, and the availability of consultants was reduced to 55% since a large proportion of their time is spent teaching or supervising junior doctors.

Beds, trolleys and treatment spaces were made available for Majors, Minors and Paediatrics separately. Due to operational differences between departments, the resources modelled in each was necessarily different. In Majors, for example, patients usually remain on trolleys throughout their stay. This is the cause of the much lamented ‘trolley queue’ which sometimes builds up in hospital corridors as patients wait to be allocated a designated trolley space in the ED. Finally, since all departments use a common x-ray machine, this was modelled as a single resource that was shared between all departments.

3.1.7. Exit points Each department ended with either the decision to admit the patient, the decision to discharge, or the patient dying. These were in most cases ‘dummy activities’ which did not correspond to specific real-world activities but were necessary within the model to manage the availability of beds for Majors patients.

The model was set up to simulate an entire year of ED activity (7 days a week, 24 h a day) with a 10-day warm-up period included to begin the simulation from dynamic steady-state conditions. Each trial contained around 40–50 runs, or however many were necessary to ensure the results fell within pre-specified confidence bands.

3.2. Verification and validation

Once the model was built, it was verified and validated in two stages, using recommendations provided by Sargent (2013). Arguably the most important level of validation involved asking clinicians to visually inspect the model to confirm that the flow of patients through the department appeared sensible, the patient flow map provided a realistic representation of the system and the logic used in the routing was correct. Several variations of the model were produced before the working group issued a statement of general acceptance of the model, inasmuch as they felt the department structure was adequately represented and valid for answering a wide range of important questions related to patient flow. The opportunity to involve clinical staff early on in the model building and validation process was invaluable in aiding them to appreciate the use of simulation as a support tool for redesign, gain ownership of the model and diligently direct how it could be developed to consider realistic and realisable alternative configurations of care.

Validation efforts then focussed on the information produced by the statistical analysis of validation data. This involved comparing the throughput and distributions of times spent by patients in different phases of system to the data for:

- Min/avg/max time in the system for patients discharged home from each of Majors, Minors and Paeds;
- Min/avg/max time in the system for patients transferred to other hospitals;
- Min/avg/max time in the system for patients referred to other healthcare sources (such as GPs);
- Min/avg/max time in the queue for triage/treatment;
- Min/avg/max length of the queue for triage/treatment; and
- % of patients discharged or admitted within the 4 h target time.

Initial validation identified some discrepancies. This was expected as some activity times in the system were based on estimates and required minimal adjustment to more accurately reflect the LoS. Some issues were also highlighted with data quality and a small pilot study found differences of the order of a few minutes between the true time of key events in patient pathways and those entered on the Symphony recording system. Of course, the simulation model can only be as accurate as the data quality allows, but as t-tests found that the KPIs listed above did not differ significantly from real-life data at the 5% level, other than the average LoS for Majors patients, the overall system was considered to be adequately modelled for the purpose of this project. The discrepancy between the true and simulated time for this care group is likely due to the effect of some real-world Majors patients being seated in an 'ambulant' area rather than requiring a bed as is assumed in the model. In future work, it would be interesting and useful to perform a time-in-motion study to obtain further data related to the emerging group of ambulant patients.

The utilisation of certain key resources was also recorded to understand which were under the most pressure. It was found that most doctors had utilisation rates around 80%, with ENPs having a rate of 70%. This was felt to correspond well to real life (combining with 'availability' values assigned earlier), and supported by staff feedback.

3.3. Results from the simulation model

With suitable comparison between the model and real-world quantitative and qualitative information, the model was used as a baseline upon which to perform 'what-if' scenario analysis. Whilst the purpose of some of the scenarios investigated was to replicate, and provide further insights into, real-world trials that had been performed at the hospital since 2012; others were designed to estimate the likely effects of further adjustments proposed to improve flow through the department (some easily realisable and others 'desirable'). Four key scenarios were examined:

1. Intra-hospital streaming (CDU)
2. Pre-hospital streaming (ambulance patients)
3. Pre-hospital streaming (GP referrals)
4. Reducing the conveyance rate of ambulance patients

The operational performance of the ED under each of the scenarios could be studied by examining the effect of several KPIs, but the most important for the client was average patient LoS (in line with the Flow Team's desire to make a genuine improvement to the underlying flow, rather than simply improve compliance with the 4 h target). For the purpose of evaluating the effect of the CDU, the analysis focussed on the LoS for discharged Majors patients staff believing that this patient group should have benefitted most from its creation (-recall that eligibility criteria is medical patients with expected LoS of 4–24 h). Furthermore, this minimised the effect of any bias resulting from changes on inpatient wards.

3.4. Intra-hospital streaming: The introduction of the CDU

The first task for the modelling team was to investigate the effect of the CDU. This was approached in a different way to traditional modelling exercises since the unit had actually been operational for 6 months before the modelling process began; however the anticipated improvement in compliance with the 4 h target in ED had not been evidenced and staff were keen to find out why. Statistical analysis of historic activity data was performed to evaluate the real-life impact of the CDU, which revealed that the establishment of the unit had only led to a 14 min decrease in LoS for patients treated in the Majors department. The team was initially surprised at this result, which was lower again than their perceived perceptions, since one would logically expect the addition of 12 extra beds to have reduced patient LoS in the ED to a greater extent. Further discussions with clinical staff however revealed that not all 12 beds were truly 'additional', but that a bed reconfiguration that coincided with the establishment of the CDU meant that 3 beds within the department were actually transferred from MAU. Hence the total number of beds actually increased by 9. Even still, hospital managers were not content with a 14 min saving and keen to discover if a larger saving could be achieved if the beds were used in a more efficient manner.

The CDU was therefore added to the model as a 'what-if' scenario, the purpose being to look at both what effect the CDU was expected to have and, later, to accurately reflect how the CDU is actually functioning, facilitating parallels to be drawn between ideal and real departments. This 'what-if' scenario was built upon the verified and validated baseline model, so there were no changes to any parameters in the model except those directly involved with the addition of the CDU. Thus the scenario essentially investigates the expected effect of the CDU, assuming that everything else remained unchanged.

However, when the structure of the CDU was added to the model according to information from managers about its proposed functioning (i.e. around 17% of Majors patients were eligible for admission to CDU and could be streamed at any point, with a bias towards earlier contact), the average waiting time of Majors patients was actually found to increase! A sensitivity analysis was performed to explore the impact of various rates of CDU admissions up to a total of 20% and found that the greatest effect on LoS was achieved when around 8–10% of patients were streamed to the unit. If more than 10% of patients were streamed, the waiting times were found to exponentially increase - explaining the reason for the increase in the average LoS described above for the 17% streaming rate. In reality, if there were more eligible CDU patients than the department could cope with, then these would be sent to other wards or processed as standard patients through the ED. Analysis of the real-life data from January - June 2013 found that 8.6% of Majors patients were actually admitted to the department, which fitted well with the recommendation of the simulation model.

The above model was named 'CDU1' and prior to changing the functioning of the CDU in the simulation, statistical analysis was conducted upon the activity levels observed in the CDU between January - June 2013. The findings of the analysis were striking and revealed that the CDU was not operating in line with the original intentions described. Most notably, the analysis found that:

- The time spent by patients in the CDU exceeded 24 h in around 30% of cases;
- Only 7% were streamed to the CDU within 2 h from the time they presented at the ED; and
- The profile of patients streamed to the CDU very closely matched those sent to the MAU.

It was clear that the CDU1 model did not reflect the reality of the department. A new model was therefore built called 'CDU2'. By combining the queues for MAU and CDU, converting the CDU to a dummy activity which was made unavailable to new eligible patients if full and routing patients to the unit later on in their stay in ED; the KPIs predicted by the model compared well to real data and revealed a great deal about the underlying problems of the way in which the CDU was functioning. It captured the use of CDU as an extension of the MAU rather than simply a fast-track stream, in a way such that CDU patients could be sent to the MAU if full (and vice versa).

This newly validated CDU2 model was then used to assess the impact of four proposed alternatives to improve the functioning of the CDU (by examining the average expected LoS of discharged Majors patients under each of the operating guidelines, which was felt to be a fair indicator of the benefit of the unit). These included:

1. *Earlier streaming* To model the CDU as the fast-track stream it was intended to be, routing percentages were altered to include a preference for routing from triage or clerking. This resulted in an improvement to indicators, with a further 7 min decrease in the average LoS for discharged Majors patients compared to the previous streaming approach. The average wait for patients to obtain a bed in the Majors department also decreased, as did LoS in the department for patients who were admitted to other wards (i.e. those not included in the model) with a total decrease of nearly 20 min compared to the baseline model.
2. *Increased routing* Previous sensitivity analysis identified 10% routing as optimum for the system. The routing percentages were increased to reflect this, and this yielded an additional 5 min decrease in LoS for Majors discharged patients.
3. *Changing the LoS profile with split beds* It was suggested that the CDU could split its 12 beds into two categories: 6 for patients staying less than 1 day and 6 for patients staying between 1 and 3 days (in recognition of the fact that in reality, many patients spend more than 24 h in the unit). This was modelled in the simulation by splitting the unit into two sections with activity times reflecting the difference in expected stays using triangular distributions with strict upper limits. The effect of this change was to further reduce LoS for both discharged Majors and Minors patients, resulting in a total decrease in LoS for discharged Majors patients from the baseline model of 28 min and a decrease in Minors LoS of 14 min from the baseline. It was therefore suggested to management that if significant effort was made to ensure length of stay in the CDU never exceeded 1 day for 50% of patients, and 3 days for the other 50%, the flow could be further improved.

The comparative results of the above trials (when implemented in succession) can be seen in Figure 4. The findings have sparked several interesting discussions between consultants

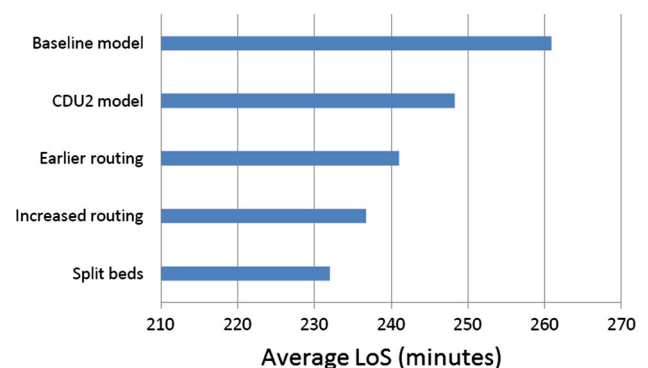


Figure 4 Compared 'what-if' scenarios for discharged Majors patients.

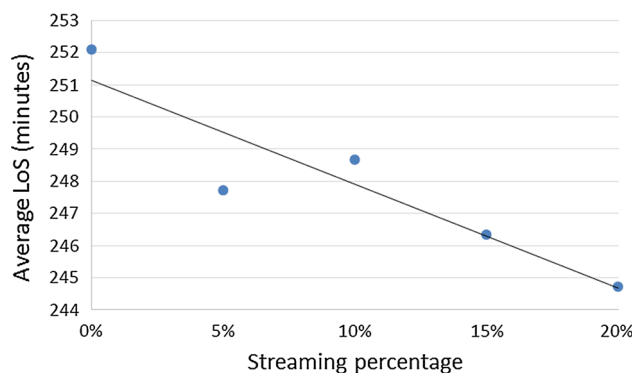


Figure 5 Change in average LoS for discharged Majors patients with increased streaming.

and hospital managers since the managers were initially surprised that the real-life impact of the unit upon ED LoS had been so small, and keen to ascertain how the unit could be reconfigured to deliver greater savings. Figure 4 shows that whilst far greater LoS reductions could be achieved if more patients were re-routed to the unit or if they were streamed earlier on in their visit, only a further 12 min reduction would be realistically achievable if **both** of these policies were implemented simultaneously. Yet, as the modelling team were keen to point out to NHS staff: this time saving is the average time saved by every discharged Majors patient, thus whilst 12 min might appear small, its cumulative effect (when amplified by the total number of patients attending the hospital on a daily basis and throughout the year), could be considerable. This fact had not been fully appreciated by NHS staff before, and thus an additional benefit to arise from the multidisciplinary weekly team meetings.

3.5. Pre-hospital streaming: Ambulance patients

A number of medical patients meeting strict clinical criteria were streamed directly from the ambulance service to the CDU for a short period last summer, bypassing the ED. Two beds within the CDU were dedicated to receiving these ambulance patients with around 20% of ambulance referrals eligible for this scheme. Limited data available from the short trial showed a total of 28 of the 65 (43%) eligible patients who were admitted to a short stay medical ward, with the CDU taking half of these. These patients typically had very short stays within the ED, which supports the model structure of external routing.

Trials were run using the CDU2 baseline model with the addition of a route available directly from the ambulance entry point to a queue for two beds in the CDU, and a number of different streamed patient proportions were explored up to a maximum of 20%. This 'queue' was a dummy activity, which allowed streaming only if beds were available. It was asserted by staff that these beds were to be used exclusively for the streaming trial and therefore they were not accessible from the wider CDU queue.

Figure 5 shows how the LoS for discharged Majors patients would be expected to decrease if the streaming of ambulance patients was increased. A significant decrease was also predicted in the length of time spent waiting to be admitted, but there was no change in Minors patients' waiting times and Paediatrics showed a very slight decreasing trend. However, for the streaming to be effective the percentage streamed had to be restricted. In fact at a 20% streaming rate, the CDU was blocked 66% of the time meaning that flow could not be established in the same way as with lower routing.

3.6. Pre-hospital streaming: GP referrals

Another streaming trial suggested for investigation was that of GP referrals, which was piloted in a very small trial at the hospital in 2012. At the point of calling the ED, suitable GP referrals were intercepted by a consultant who aimed to redirect or reschedule attendances. This model was structured slightly differently to that in Section 3.5 which streamed patients directly from their entry point. Since there was little information available about the numbers of suitable patients for the GP streaming, patients were streamed from the call handling point in the model instead. Patients were assigned a label that indicated their suitability according to the probability profile distribution 'GP streaming' upon entry to the system. This label was then used to route patients from call handling to the CDU. A label of 'eligible' did not guarantee that the patient would be streamed, only that these patients would be subject to a different set of routing percentages specified at the call handling work station. With little information on the eligibility and streaming of patients for this scheme, the accuracy of the results output from this investigation relies heavily upon the assumption that the estimates provided by staff were realistic, but a 10% eligibility of patients with a total of 40% of those eligible being routed to the CDU saw the largest reduction in the LoS for discharged Majors patients. The extra resource required from clinicians to intercept each GP referral was thought to be minimal and therefore not directly built into this model, but could be readily added if appropriate time estimates are provided in future.

3.7. Reducing ambulance conveyances

Diefenbach and Kozan (2011) have previously highlighted that the ED must treat all patients that present and therefore the main method of reducing the flow of arrivals is to divert ambulances before they arrive. Such diversions are initiated at the hospital in extreme overcrowding situations, but the client wanted to gain an appreciation of the improvement in KPIs that could result from an extension of this initiative: specifically asking ambulance crews to continually limit the number of patients conveyed to the department as far as would be considered clinically safe. The pre-hospital ambulance

streaming trial demonstrated the merit of collaboration between services that feed in to hospital EDs, but a reduction in ambulance conveyances would arguably require a significant culture change within the ambulance service, supported by the relevant rules and guidance. This is discussed in more detail in Snooks *et al.* (2002) and reviewed again in Snooks *et al.* (2004). The latter study found that increased clinical risk added to the complexity of conveyance reduction as many studies show a significant minority of those not conveyed at the risk of deterioration.

Whilst significant costs and risks would be associated with launching a trial to investigate the effects of reducing conveyance rates in the real world, predictions of improvements that could be gained from altering the demand profile of ambulance referrals were relatively straightforward to obtain with the simulation, allowing exploration of bias against lower category resulting from more of these less severe patients treated at scene. Discussion with managers of the local Health Board's ambulance service revealed that they were aiming to reduce patient conveyance rates by over 10%, from around 70 to 60%. Thus the model was used to quickly consider the effects of decreased conveyance rates on the system, revealing that successful realisation of this goal would reduce the average amount of time spent in ED by Majors patients by an average of 10 min.

3.8. Impact and learning

The actions that have followed since presenting clinicians with the data underpinning the functioning of their department have been striking and exemplify the proper use of OR: whilst clinical staff admitted that the findings of the CDU1 model seemed accurate and realistic without hesitation; they did not begin to make physical changes within the ED until the model and recommendations were presented. At present, a team of consultants are working together to redesign the unit. They have reallocated an advanced nurse practitioner to pull eligible patients across to the CDU at the point of triage and are looking to manage patients more carefully in an attempt to realign the CDU back to its original intention. Furthermore, the hospital has recently invested further resources in GP call handling and a business case for more staff has been compiled based on the work presented.

The positive outputs of the ambulance conveyance study together with other research works highlighting the benefits of reduced conveyance rates (see O'Cathain *et al.*, 2016; Snooks *et al.*, 2013), have additionally incited the launch of a Physician Response Unit pilot within the Health Board. The pilot involves sending ED physicians out with ambulance crews on set days of the week to actively encourage the treatment of patients at the scene of the incident, wherever it is safe and appropriate to do so. To date, the trial has been successful with a statistically significant drop that is found in the conveyance rates on the days on which Physicians Response Unit has been active.

Whilst discussions regarding the best way to implement further initiatives at the hospital are ongoing, this project has made a considerable impact upon the mindset of NHS staff at the department who now value the benefit of exploring simulation methods before changes are implemented. The General Manager of Unscheduled Care recently commended the work, stating that "the outputs from the project exceeded expectations, in that it provided a level of understanding which went on beyond what was anticipated...This has been an excellent way of making operational teams aware of the potential for using new improvement methods such as using simulation as a tool to design future services." Clinical staff in the Health Board are also increasingly becoming more and more convinced of the value of this approach and in agreement that modelling is crucial to planning future services in the right way.

4. Concluding remarks

DES is an effective tool for patient flow modelling and this study has illustrated that it can be expected to have an important role in designing future ED services. The simulation model developed in this study has enabled managers to test out the feasibility of proposed initiatives prior to implementation and is currently being used to support the implementation of specific reconfigurations.

In order to provide the reader with a realistic example of timescales and software required to set up a simulation study and to bring the research into a real-world implementation, a timeline of this particular case study is provided in Figure 1 and a list of software in Appendix A. Whilst it is by no means necessary for project managers of future studies to use identical software packages, the timeline should provide a useful indication of the typical timeframes required to complete each stage of a successful ED simulation study. Whatever software packages used, some form of stochastic modelling is strongly advocated to evaluate potential patient flow strategies since deterministic spreadsheet models that do not adequately account for randomness can be liable of recommending insufficient staffing resources. Moreover, our experience of setting up the simulation study agrees with that of Bowers *et al.* (2009), in that its ability to bring stakeholders together to constructively debate the complexities of the ED system proved to be invaluable. Our principal advice to project managers of future is that similar studies are thus to set up a meaningful dialogue between the mathematical modellers, data analysts, managers and clinicians from the outset of the project and to network regularly throughout. Applying improvement techniques requires expertise. Hence any organisation intending to do this work should initially identify several clinical champions in addition to a larger cohort of energised workers who are capable of overseeing the operations of the group and ensuring that the changes are correctly implemented in the workplace. At minimum, the stakeholder

team should include a project champion, key stakeholders and clinicians; and the roles of a simulation modeller, facilitator and recorder should be appropriately assigned between other team members. To certify that the model is used as anticipated, it is not only imperative for the healthcare workers to ensure that the modellers have a detailed understanding of all data caveats and hospital process; but also for the modelling team to ensure that the healthcare workers are fully aware of the model limitations.

In future research it would be interesting to investigate both more detailed and wider modelling of the department. It is hoped that improvements in data collection and database connections (i.e. between servers that hold important information about patients' pathways linking primary and secondary care) will soon simplify this process. If data advancements allow, it would also be useful to integrate the model in a multi-level framework, taking account of both upstream and downstream services to facilitate a more cohesive insight into cause and effect in the system.

As an appetite for modelling continues to grow at the Health Board, there is significant interest in extending this model to incorporate some of the above suggestions and carefully monitor the impact of each implemented change as they arise. In response to the increased desire of management and clinical staff at the Health Board to see mathematical modelling become an integral part of NHS design and delivery, a fulltime modeller has recently been appointed who hopes to oversee this work in addition to pioneering other novel simulation techniques in healthcare.

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Appendix A: List of software

- **SIMUL8** A simulation tool that allows the user to create a virtual model of any department that processes discrete entities at discrete times. Basic features of the model can be built via a simple graphical interface and more advanced features can be added using SIMUL8's own simulation language, Visual Logic. It's design allows communication with other software packages, so parameters and results can be simply imported/exported. Popular alternatives to SIMUL8 include AnyLogic, Witness, Simio, ARENA, ProModel, Flexsim and Scenario Generator (for strategic planning for health and social care). Further guidance on selecting the most appropriate package is provided in Azadeh et al (2010).
- **R** Open-source data analysis software in which the analysis is performed via scripts and functions written in the R programming language. Not only can the software be downloaded for free, but the source code required to perform the data analysis can be recycled from other projects and directly modified by the user. In this study, R was used for distribution fitting but it can also be used for general statistical analysis, data visualisation and predictive modelling. Alternative proprietary software include SAS, IBM SPSS and Stata.
- **Microsoft Excel** Excel is widely available within many organisations and most NHS staff have a basic understanding of the software. Excel does contain a limited number of add-ins that can be used to perform data analysis, but its functionality remains very basic in comparison to the packages listed above.

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